

SUZHE XU

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EDUCATION

Huaqiao University

Sep. 2023 – Jun. 2027 (expected)

Undergraduate, Artificial Intelligence, School of Computer Science and Technology

Comprehensive evaluation rank (2023–2025): 1/40; academic rank: 4/40.

GPA: 4.09/5.00 (84.77/100).

ACHIEVEMENTS

National Scholarship, China.

First Prize (Provincial Level), China Undergraduate Mathematical Contest in Modeling (CUMCM).

Completed project, National Undergraduate Innovation and Entrepreneurship Training Program (China).

RESEARCH INTERESTS

Efficient generative modeling: native few-step generation, diffusion distillation, video inverse problems, and streaming world models; training stability and multimodal post-training.

PUBLICATIONS AND MANUSCRIPTS

1. Weimin Bai, **Suzhe Xu**, Yiwei Ren, Jinhua Hao, Ming Sun, Wenzheng Chen, and He Sun. **InstantViR: Real-Time Video Inverse Problem Solver with Distilled Diffusion Prior**. *CVPR*, 2026. **Highlight**. Co-project lead and presenter. [\[Paper\]](#) [\[Project\]](#) [\[Presentation\]](#)
2. **Suzhe Xu**^{*}, Weimin Bai^{*}, Jinhua Hao, Ming Sun, Wenzheng Chen, and He Sun. **InstantViR++: Improved Posterior Distillation for Real-Time Video Inverse Problem Solving**. *Submitted to IEEE Transactions on Multimedia*, 2026. ^{*}Equal contribution.
3. **KAT-Coder-V2 Technical Report**. arXiv:2603.27703, 2026. [\[Report\]](#)
Contributions: execution-based WebCoding data curation, supervised fine-tuning, and long-context continued pretraining.

RESEARCH EXPERIENCE

Peking University

Jun. 2025 – Present

Visiting Student / Research Assistant, Wangxuan Institute of Computer Technology

Computational Imaging Lab; advised by He Sun and Wenzheng Chen.

InstantViR and InstantViR++

Real-time video inverse problems

- Co-led InstantViR from method design to implementation, experiments, and CVPR presentation. Amortized iterative diffusion posterior sampling into a causal one-step solver for video deblurring, inpainting, and super-resolution.
- Developed the distillation pipeline, causal autoregressive solver, and LeanVAE integration. The published system exceeds **35 FPS** with up to **100 $\times$ speedup** over iterative solvers, without external paired clean/degraded training videos.
- Investigated cross-VAE posterior distillation for InstantViR++ with Kuaishou collaborators, using bidirectional latent conversion and cyclic loss weighting to balance measurement consistency and generative supervision, with controlled comparisons.

RESEARCH EXPERIENCE (CONTINUED)

Streaming World Models for Autonomous Driving

Jun. 2026 – Sep. 2026

Industry-collaborative research project

Peking University

- Developed a bidirectional-teacher → causal-teacher → consistency-distilled student pipeline for Waymo videos, enabling four-step block-autoregressive generation. Designed a clean KV cache to condition subsequent blocks on committed clean history.
- Implemented differentiable self-forcing rollouts and 25/49-frame training. Used ablations to diagnose exposure bias and training–inference mismatch; extended the pipeline to multi-view generation.
- *Prometheus: Controllable Video World Models for Autonomous Driving*. Technical report in preparation; working title.

Multi-Agent World Model Post-Training and Evaluation

Ongoing

Research project

Peking University

- Adapted Cosmos3-Edge for native dual-view, action-conditioned multi-agent generation, integrating independent action tokens and Gamma Simplex four-axis positional encoding to represent agent identity, time, and space. Completed staged training with a 93-to-189-frame curriculum on **16 A100 GPUs across two nodes**.
- Established protocol-matched comparisons with the Gamma teacher by aligning frame rates, prediction horizons, and sampling budgets; analyzed discrepancies between validation loss and visual fidelity, including persistent material and geometric errors.

RedNote (Xiaohongshu), Dots

Sep. 2026 – Present

Multimodal Research Intern

- **Native few-step language generation:** Establish FLM cross-entropy/MSE baselines and adapt MeanFlow and simplified consistency training. Evaluate 1/2/4-step generation with Gen PPL, MAUVE, repetition rates, and samples to study objective–quality discrepancies and quality–diversity–compute trade-offs.
- **Distribution-matching distillation:** Investigate brightness degradation and structural loss in JiT-based image distillation using exact finite-support scores, direct pixel optimization, and generator updates. Disentangle score estimation, optimization, and parameterization effects; observe amplified global color/brightness updates under network parameterization in controlled settings.
- Designed and managed parallel 8-GPU experiments with peak aggregate usage exceeding **100 GPUs**, coordinating tracking, checkpointing, and evaluation for controlled quality–efficiency comparisons.

ADDITIONAL INDUSTRY EXPERIENCE

Kuaishou

Jan. 2026 – Jun. 2026

Algorithm Research Intern, Frontier Agent Algorithms

- Developed **InstantViR++** during the internship with Jinhua Hao (Kling AI) and Peking University using Kuaishou compute resources.
- Built a WebGen-Bench-based verification pipeline combining LLM-generated tests, Selenium/Chrome, and WebVoyager UI agents. Curated **10k+** web-tool/game examples for SFT and extended evaluation to 3D scenarios.
- Built a CPT pipeline spanning **3,600+ repositories** and **187,858 examples**, with Issue–PR to FIM/unified-diff transformations and 64K/256K packing. Trained and evaluated Qwen3.5-35B-A3B at 64K context using Megatron-Bridge on **32 H800 GPUs**.

SKILLS AND LANGUAGES

Technical Skills: PyTorch, distributed GPU training, FSDP, LoRA, Megatron-Bridge, TensorBoard, Git, LaTeX.

Languages: Mandarin Chinese; English (IELTS 7.5 overall); French (B1, self-assessed).